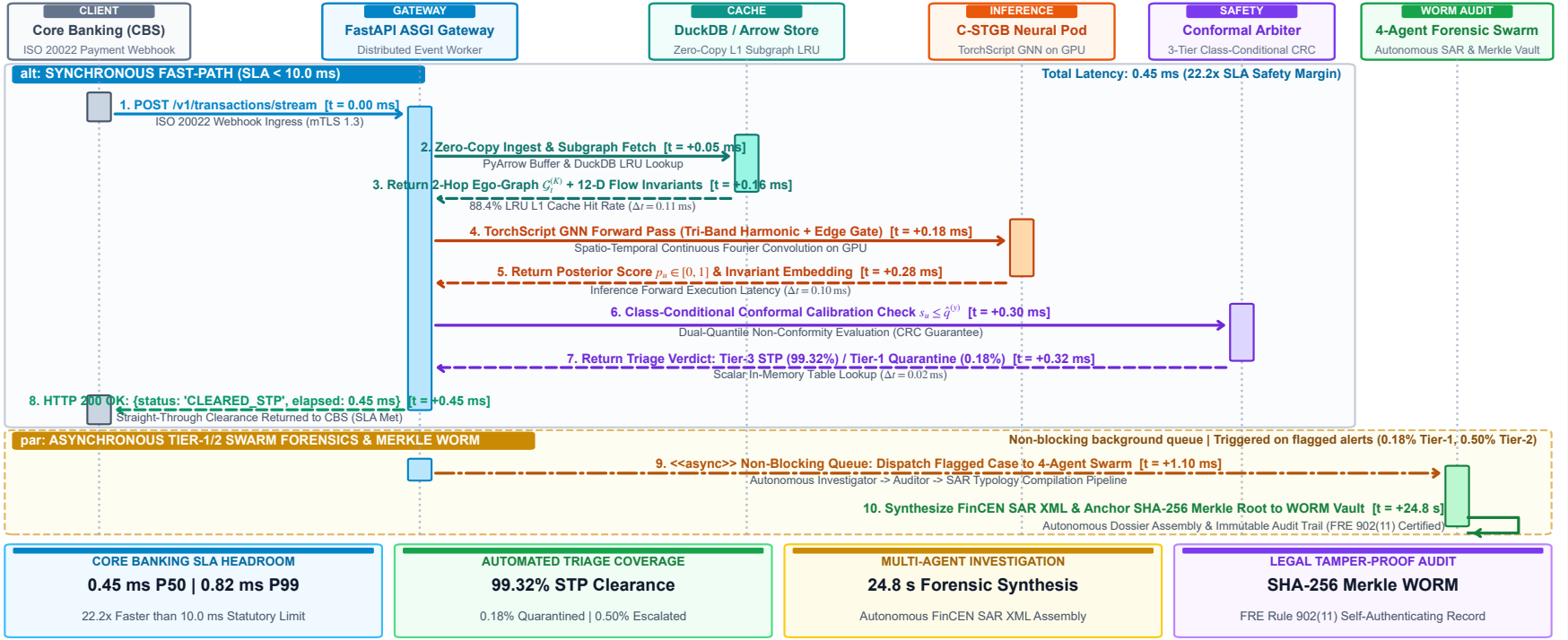
Figure 8.2: Real-Time Operational UML Sequence Lifeline & Microsecond Execution Trace

Deterministic Microservice Pipeline: ISO 20022 Ingress → Vectorized LRU Cache → C-STGB Forward Pass → Conformal Triage → Merkle WORM

FAST-PATH: 0.45 ms (SLA < 10.0 ms MET)



CORE BANKING SLA HEADROOM

0.45 ms P50 | 0.82 ms P99

22.2x Faster than 10.0 ms Statutory Limit

AUTOMATED TRIAGE COVERAGE

99.32% STP Clearance

0.18% Quarantined | 0.50% Escalated

MULTI-AGENT INVESTIGATION

24.8 s Forensic Synthesis

Autonomous FinCEN SAR XML Assembly

LEGAL TAMPER-PROOF AUDIT

SHA-256 Merkle WORM

FRE Rule 902(11) Self-Authenticating Record